# Calcurion Maths Worksheets

A-Level Further Mathematics

**Topic:** Further Pure 1 – Further Complex Numbers

Medium & Hard • 25 Questions

Scan for free resources

## Section A – Medium (Q1–15)

**1.** Describe the locus $|z - 2 - 3i| = 4$. **[2]**

**2.** Describe the locus $\arg(z - 1) = \frac{\pi}{3}$. **[2]**

**3.** Find the Cartesian equation of $|z - 1| = 2|z + i|$. **[4]**

**4.** Describe the transformation $w = z + 2 - i$. **[2]**

**5.** Describe the transformation $w = iz$. **[2]**

**6.** Describe the transformation $w = \bar{z}$. **[2]**

**7.** Find the image of the line $\mathrm{Re}(z) = 1$ under $w = z^2$. **[4]**

**8.** Find the image of $|z| = 1$ under $w = \frac{1}{z}$. **[3]**

**9.** Find the fixed points of $w = z^2$. **[3]**

**10.** Describe the locus $\mathrm{Re}(z^2) = 1$. **[4]**

**11.** Find $w$ if $z = 1 + i$ and $w = \frac{z-i}{z+i}$. **[3]**

**12.** Show that $w = \frac{z-1}{z+1}$ maps the unit circle to a line. **[4]**

**13.** Find the locus of $z$ such that $\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{2}$. **[4]**

**14.** Describe $w = 2z$ geometrically. **[2]**

**15.** Find the image of the half-plane $\mathrm{Im}(z) > 0$ under $w = \bar{z}$. **[2]**

## Section B – Hard (Q16–25)

**16.** Find the image of the circle $|z - 1| = 1$ under $w = \frac{1}{z}$. **[5]**

**17.** The transformation $w = z + \frac{1}{z}$ is applied to $|z| = 2$. Find the Cartesian equation of the image. **[5]**

**18.** Find the locus $\left|\frac{z-1}{z+1}\right| = 2$. **[4]**

**19.** Show that $w = \frac{az+b}{cz+d}$ (Möbius) maps circles/lines to circles/lines. **[4]**

**20.** Find the values of $z$ for which $w = z^2$ maps to purely imaginary $w$. **[4]**

**21.** Describe the region $|z - 1| < |z + 1|$ and its image under $w = iz$. **[4]**

**22.** Find the inverse transformation of $w = \frac{z-i}{z+i}$. **[4]**

**23.** Sketch the locus $\arg(z-2) - \arg(z+2) = \frac{\pi}{4}$. **[5]**

**24.** Find the image of the strip $0 \leq \mathrm{Re}(z) \leq 1$ under $w = e^z$. **[5]**

**25.** Solve for $z$ if $w = z + \frac{1}{z}$ and $w = 2i$. **[4]**

*End of Worksheet* • *Show all working clearly.*